11/8/26

5. Develop a customized ping command to test the server Connectivity

Aim :

To develop a customized Ping command in Python that checks the connectivity between a client and a server by sending ICMP Echo request packets and displaying the response time and status.

Steps :

1. Install Python
2. Install VS Code
3. Create Project Folder
4. Execute the Program
5. Enter website

Result : Thus the customized command developed Successfully.

gj.py > ...

```python
import subprocess
import platform
print("="*50)
print("Customized Ping Utility")
print("="*50)
host = input("Enter Website or IP Address: ")
system = platform.system()
if system == "Windows":
    command = ["ping", "-n", "4", host]
else:
    command = ["ping", "-c", "4", host]
try:
    result = subprocess.run(command, capture_output=True, text=True)
    print("\nPing Results\n")
    print(result.stdout)
    if result.returncode == 0:
        print("Server Status : Reachable")
    else:
        print("Server Status : Unreachable")
except Exception as e:
    print("Error:", e)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Computer Network>  d:; cd 'd:\Computer Network'; & 'C:\Users\REC\AppData'
in32-x64\bundled\libs\debugpy\launcher' '54199' '--' 'd:\Computer Network\gj.py
==================================================
Customized Ping Utility
==================================================
Enter Website or IP Address: google.com

Ping Results


Pinging google.com [142.251.221.142] with 32 bytes of data:
Reply from 142.251.221.142: bytes=32 time=29ms TTL=120
Reply from 142.251.221.142: bytes=32 time=20ms TTL=120
Reply from 142.251.221.142: bytes=32 time=53ms TTL=120
Reply from 142.251.221.142: bytes=32 time=28ms TTL=120

Ping statistics for 142.251.221.142:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 20ms, Maximum = 53ms, Average = 32ms

Server Status : Reachable
```